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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I Semester B. E. Examinations April-2021

Common to All Branches

ELEMENTS OF MECHANICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of steam tables is permitted.

PART-A

1	1.1	Difference between superheated temperature and saturation temperature is called as _____.	01
	1.2	_____ is used for high head and low discharge of water.	01
	1.3	The amount of heat required converting *one* kg of water at saturation temperature into dry saturated steam at same temperature and pressure is known as _____.	01
	1.4	A petrol engine works on _____ cycle.	01
	1.5	What is the value of saturation temperature of water at 15 bar pressure?	01
	1.6	Both heat and work are _____ functions.	01
	1.7	Which part of the IC engine stores energy during power stroke and uses it during other strokes?	01
	1.8	Compression ratio of petrol engine as compared to diesel engine is _____.	01
	1.9	Following are the readings of a single cylinder four stroke petrol engine. Load on the brake drum $= 40kgf$, spring balance reading $= 5kgf$, and diameter of the brake drum $= 120cm$. Find torque on the brake drum.	02
	1.10	Arbor is one of the parts of _____ machine.	01
	1.11	The process of uniformly reducing the diameter over the length of cylindrical work piece by using lathe is called as _____ process.	01
	1.12	The ratio of oxygen to acetylene in neutral flame is _____.	01
	1.13	In _____ type of welding, the two work pieces to be welded are heated to molten state.	01
	1.14	For a simple gear train, the velocity ratio (in terms of number if teeth) is defined as _____.	01
	1.15	_____ gear is used to transmit the power from one shaft to another which are parallel.	01
	1.16	What is reaming operation?	01
	1.17	In open belt drive, both the pulleys are rotating in _____ direction.	01

	1.18	Hard solder are made of an _____ alloy.	01
	1.19	In _____ type of milling, the direction of rotation of cutter and feed are opposite.	01

PART-B

2	a	Explain the following with simple sketches: i) Zeroth law of thermodynamics ii) Kelvin Plank statement.	06
	b	Define the following: i) Dryness fraction of steam ii) Enthalpy of super-heated steam iii) Specific volume of dry saturated steam iv) Sensible heat.	04
	c	Find the enthalpy of $1kg$ of steam at $12bar$ when: i) Steam is 22% wet ii) Steam is dry saturated iii) Super-heated to 250^0C. Assume the specific heat of the superheated steam as $2.25kJ/kg/K$.	06
3	a	Explain with a neat sketch, a pelton wheel.	06
	b	Explain open cycle gas turbine with a neat sketch.	06
	c	List: i) Desirable properties if good refrigerant ii) Different types of refrigerant.	04
		OR	
4	a	With a neat sketch, explain the working of vapor compression refrigeration.	08
	b	Explain: i) Refrigeration effect ii) Ton of refrigeration.	04
	c	Differentiate between open and closed cycle gas turbine.	04
5	a	With the P-V diagram, explain working of four stroke diesel engine.	08
	b	A four cylinder four stroke petrol engine develops indicated power of $15kW$ at $1000rpm$. The indicated mean effective pressure is $0.55MPa$. Calculate the bore and stroke of the piston, if the length of stroke is 1.5 times the bore.	08
		OR	
6	a	Differentiate between two stroke and four stroke engine.	04
	b	Explain how internal combustion engines are classified?	04

	c	The following observations were obtained during a trial on a four stroke diesel engine. — Cylinder diameter = $25cm$ — Stroke of the piston = $40cm$ — Crank shaft speed = $250rpm$ — Brake load = $70kg$ — Brake drum diameter = $2m$ — Mean effective pressure = $6\ bar$ — Diesel oil consumption = $0.1m^3/min$ — Specific gravity of diesel = 0.78 — Calorific value of diesel = $43900kJ/kg$ Find: i) Brake power ii) Indicate power iii) Mechanical efficiency iv) Friction power v) Brake thermal efficiency vi) Indicated thermal efficiency.	08
7	a	How do you specify a lathe?	04
	b	Explain with a neat sketch, following operation: i) Thread cutting ii) Boring iii) Counter sinking iv) Tapping.	08
	c	Explain the following with a neat sketch: i) Up milling ii) Angular milling.	04
8	a	Classify welding process. Explain with a neat sketch, working principle of arc welding.	06
	b	Draw neat sketches of open and cross belt drive and differentiate them.	04
	c	Write a note on CNC machines. What are its advantages and disadvantages over conventional machines?	06